

Travel & Expense Budget Allocation Optimization

A public, anonymized case study in financial planning, index design, and decision support

MY ROLE

Finance Intern

BUSINESS CONTEXT

**Fortune 50 home improvement
retailer**

CORE SKILLS

**FP&A | Data modeling | Executive
communication**

THE CHALLENGE

A legacy, cost-focused budgeting approach did not fully capture how customer density, cost of living, travel distance, and workforce mix changed T&E needs across markets.

~4%

Higher cost pressure

Observed across the most heavily staffed market group after accounting for local operating conditions.

~10%

Higher travel difficulty

Observed among the highest-staffed markets, showing that headcount alone did not explain travel needs.

10x+

Range in spend per head

The wide spread highlighted how workforce composition and geography can materially change allocation needs.

Public portfolio note: Market names, employee names, formulas, and internal dollar data have been removed. Findings are rounded and presented only to demonstrate the analytical approach.

Building a multi-factor planning signal

I designed a repeatable framework that translated operating differences into clearer budget-planning inputs.

INPUTS

Four drivers of T&E demand

- 01 Customer density**
Concentration of demand and service activity
- 02 Cost of living**
Local price pressure affecting trip costs
- 03 Travel distance**
Average separation across service geography
- 04 Workforce mix**
Role-level needs and headcount composition

MODELING WORKFLOW

- 1 Audit**
Compared the legacy planning signal with actual market characteristics.
- 2 Build**
Created separate cost and travel-difficulty indexes.
- 3 Stress-test**
Checked distributions, outliers, and ranking sensitivity.
- 4 Compare**
Flagged markets with potential over- or under-allocation.
- 5 Operationalize**
Structured outputs for integration into existing planning templates.

ANALYTICAL JUDGMENT

Distance data for non-contiguous geographies created artificial spikes. I evaluated log transformation, cube-root transformation, and capping to reduce outlier influence without flattening meaningful differences.

Key principle: preserve the business signal, not just improve the distribution.

What changed - and why it mattered

01 Headcount was not enough

Markets with similar staffing could have very different travel burdens and local cost structures.

02 Cost and distance told different stories

Separating the signals made it easier to see whether local prices or service geography drove the need.

03 The legacy index could misdirect funds

The multi-factor view surfaced potential over- and under-allocation hidden by a single cost measure.

DELIVERABLE

A repeatable allocation framework

- Two complementary planning indexes: cost and travel difficulty
- A transparent comparison to the existing planning signal
- A process to refresh inputs and flag allocation exceptions each cycle
- Outputs designed for adoption in existing planning templates

BUSINESS IMPACT

Better allocation logic

Connected T&E funding to the operating characteristics that create demand.

Stronger planning conversations

Gave finance and business partners a clearer rationale for market-level adjustments.

More sustainable process

Reduced dependence on one-off judgment by creating a repeatable analytical workflow.

WHAT I LEARNED

The strongest model is not the most complex one - it is the one that balances analytical rigor, business judgment, and ease of adoption.